

COMP 311 — Fall 2026 Course Calendar

Week	Date	Day	Unit	Week Theme	Topic / Class Plan	Guiding Question	H&P Reading
1	8/17/2026	Monday	Unit 1: Programming the Machine	Programs Below the Program	Course introduction; abstraction layers; first RISC-V program	What happens below a high-level program?	1.1–1.4; 2.1
1	8/19/2026	Wednesday	Unit 1: Programming the Machine	Programs Below the Program	Registers, operands, and arithmetic instructions	What machine state can an instruction access?	2.2–2.3
2	8/24/2026	Monday	Unit 1: Programming the Machine	Representing and Manipulating Values	Binary, hexadecimal, signed and unsigned integers	How does the machine represent values?	2.4
2	8/26/2026	Wednesday	Unit 1: Programming the Machine	Representing and Manipulating Values	Addition, subtraction, overflow, logical operations, and shifts	How do bit patterns become useful computations?	2.6; 3.2
3	8/31/2026	Monday	Unit 1: Programming the Machine	Data and Control Flow	Memory, addresses, lw, sw, and arrays	How does a program access data that does not fit in registers?	2.3; 2.14
3	9/2/2026	Wednesday	Unit 1: Programming the Machine	Data and Control Flow	Quiz 1; comparisons, branches, and loop introduction	How does the machine choose which instruction executes next?	Review 2.2–2.6; begin 2.7
4	9/7/2026	Monday	No Class	Assembly Synthesis	Labor Day — no class		

4	9/9/2026	Wednesday	Unit 1: Programming the Machine	Assembly Synthesis	Conditionals, loops, arrays, and translation practice	How do high-level control structures map to RISC-V?	2.3; 2.7; 2.14
5	9/14/2026	Monday	Unit 1: Programming the Machine	Building Larger Assembly Programs	Procedures, arguments, return values, jal, and jalr	How can one assembly routine call another?	2.8
5	9/16/2026	Wednesday	Unit 1: Programming the Machine	Building Larger Assembly Programs	Quiz 2; stack frames and recursion introduction	How does the machine preserve state across nested calls?	Review 2.3, 2.7, 2.8, 2.14
6	9/21/2026	Monday	No Class	From Assembly to Bits	Well-Being Day — no class		
6	9/23/2026	Wednesday	Unit 1: Programming the Machine	From Assembly to Bits	Recursion synthesis; machine-code encoding and RISC-V instruction formats	How are procedure behavior and instructions represented in bits?	2.5; 2.8
7	9/28/2026	Monday	Unit 1: Programming the Machine	Completing the Programmer-Visible Machine	Wide immediates, addresses, assembly, linking, and loading	How does symbolic assembly become an executable program?	2.10; 2.12
7	9/30/2026	Wednesday	Unit 1: Programming the Machine	Completing the Programmer-Visible Machine	Quiz 3; ISA synthesis and transition to hardware	What hardware capabilities have our programs assumed?	Review 2.5, 2.8, 2.10, 2.12
8	10/5/2026	Monday	Unit 2: Inside the Processor	Computing with Logic	Gates, truth tables, Boolean expressions, and abstraction	How can physical circuits represent logical decisions?	Appendix A. 1–A.2 / zyBooks 7.1–7.2

8	10/7/2026	Wednesday	Unit 2: Inside the Processor	Computing with Logic	Combinational logic, muxes, decoders, and reusable components	How can simple gates become useful hardware blocks?	Appendix A. 3 / zyBooks 7.3
9	10/12/2026	Monday	Unit 2: Inside the Processor	Arithmetic and State	Adders, subtraction, comparison, and ALU construction	How can one circuit support many RISC-V operations?	Appendix A. 5–A.6 / zyBooks 7.5–7.6; 3.2
9	10/14/2026	Wednesday	Unit 2: Inside the Processor	Arithmetic and State	Clocks, state, flip-flops, and registers	How does hardware remember values over time?	Appendix A. 7–A.8 / zyBooks 7.7–7.8
10	10/19/2026	Monday	Unit 2: Inside the Processor	Storage Components and Timing	Register files, memory components, and timing conventions	How does the processor read and update architectural state?	Appendix A. 9–A.11 / zyBooks 7.9–7.11; 4.2
10	10/21/2026	Wednesday	Unit 2: Inside the Processor	Storage Components and Timing	Quiz 4; register-transfer view and datapath preview	How can we describe an instruction as data moving between components?	Review Appendix A; preview 4.1–4.2
11	10/26/2026	Monday	Unit 2: Inside the Processor	Constructing the Datapath	Datapath for R-type and immediate instructions	Where does information travel during an instruction?	4.1–4.3
11	10/28/2026	Wednesday	Unit 2: Inside the Processor	Constructing the Datapath	Extending the datapath for loads and stores	How can one datapath support registers and memory?	4.3
12	11/2/2026	Monday	Unit 2: Inside the Processor	Completing the Processor	Branches, jumps, immediate generation, and control	How does control select the correct datapath behavior?	4.3–4.4

12	11/4/2026	Wednesday	Unit 2: Inside the Processor	Completing the Processor	Quiz 5; complete processor, critical path, and introduction to pipelining	The processor works—but how can we make it faster?	Review 4.1–4.4; 1.6; begin 4.6
13	11/9/2026	Monday	Unit 3: Making Computers Fast	Pipelining	Performance equation, single-cycle limitations, pipeline motivation, and five-stage pipeline	How can the processor complete instructions more frequently?	1.6; 4.6–4.7
13	11/11/2026	Wednesday	Unit 3: Making Computers Fast	Pipelining	Pipeline datapath, data hazards, and forwarding	What happens when an instruction needs an unfinished result?	4.7–4.8
14	11/16/2026	Monday	Unit 3: Making Computers Fast	Pipeline Hazards	Stalls, load-use hazards, control hazards, and flushing	How does the pipeline handle dependencies and uncertain control flow?	4.8–4.9
14	11/18/2026	Wednesday	Unit 3: Making Computers Fast	Pipeline Hazards	Quiz 6; pipeline synthesis and transition to memory hierarchy	What limits performance after we pipeline the processor?	Review 4.6–4.9
15	11/23/2026	Monday	Unit 3: Making Computers Fast	The Memory Wall	Locality, memory hierarchy, and cache fundamentals	Why can't the processor simply access main memory every cycle?	5.1–5.3
15	11/25/2026	Wednesday	No Class	The Memory Wall	Thanksgiving recess — no class		
16	11/30/2026	Monday	Unit 3: Making Computers Fast	Performance and Course Synthesis	Cache address breakdown, hits and misses, AMAT; virtual memory or parallelism overview	How does memory behavior affect total program performance?	5.3–5.4; selected 5.7 or 6.1–6.6

16	12/2/2026	Wednesday	Unit 3: Making Computers Fast	Performance and Course Synthesis	Whole-course synthesis and final-exam review	How does a program become hardware activity, and what determines its speed?	Review 2.1–2.14, 4.1–4.9, 5.1–5.4
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Quiz	Assignment / Lab Alignment
	WA 1 released: RISC-V basics, registers, arithmetic, representation
	WA 1 due
Quiz 1: registers, arithmetic, binary/hex, signed	WA 2 released: memory, arrays, branches, loops, basic procedures

	WA 2 due
Quiz 2: memory, arrays, loads/store s	WA 3 released: procedures, stack, recursion, encoding
	WA 3 due
Quiz 3: stack behavior, recursion, instruction	

	WA 4 released: Boolean/combinational /sequential logic; Lab 1 released: Combinational Logic +
	WA 4 due; Lab 2 released: Register File
Quiz 4: gates, Boolean logic, muxes	Lab 1 due; WA 5 released: datapath + control tracing
	Lab 3 released: Datapath
	WA 5 due; Lab 2 due
	Lab 4 released: Complete Single-Cycle CPU

Quiz 5: single-cycle datapath, control signals	WA 6 released: pipelining + hazards
	Lab 3 due
	WA 6 due
Quiz 6: pipeline stages, performan ce	Lab 4 due; Lab 5 released: Cache / Performance Analysis

	Lab 5 due
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